Problem

Problem · Superposing two oscillations. In physics, a simple voltage or sound can be written $t \mapsto A\cos(\omega t + \varphi)$. We show here that a sum $a\cos x + b\sin x$ can be put in this form. We study $g(x) = \sqrt{3}\cos x + \sin x$.

1. Write $\sqrt{3} - i$ in exponential form.
2. Prove that $g(x) = \mathrm{Re}\left(\left(\sqrt{3} - i\right)\mathrm{e}^{ix}\right)$, then that $g(x) = 2\cos\left(x - \frac{\pi}{6}\right)$.
3. Deduce the maximum and the minimum of g on $\mathbb{R}$, and the real numbers in $[0, 2\pi)$ where they are reached.
4. Solve in $[0, 2\pi)$ the equation $g(x) = \sqrt{2}$.
5. Generalise: let a and b be two real numbers, not both zero. Prove that there exist $R > 0$ and a real number φ such that, for every real number x, $a\cos x + b\sin x = R\cos(x - \varphi)$. Express R in terms of a and b.